MILITARY SPECIFICATION

FLASK, ERLENMEYER, GRADUATED, PLASTIC

This limited coordination Military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-F-36747(DSA-DM). It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of autoclavable plastic and graduated Erlenmeyer flask with narrow mouth and reinforced lip, suitable for use in laboratory procedures.

1.2 Classification. Erlenmeyer flasks covered by this specification shall be of the following sizes, as specified (see 6.2):

Size 1 - 50 milliliters (ml.) capacity.
Size 2 - 250 milliliters (ml.) capacity.
Size 3 -1000 milliliters (ml.) capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-393	Plastic Molding Material, Polycarbonate, Injection Extrusion.
PPP-P-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.
ZZ-S-751	Stopper, Bottle, Rubber.

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

FSC 6640

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The flask shall be fabricated from polycarbonate resin conforming to L-P-393, Form 1.

* 3.2 Construction. The flask shall be transparent and shall be of one-piece construction, having conical sides that terminate in a narrow cylindrical mouth having a flared lip for reinforcement. The bottom base of the flask may be slightly concave. The flask shall conform to figure I for style, design and dimensions. Graduation markings shall be raised on the outside surface with capacity lines and numbering as indicated in figure I. The raised graduation markings shall be permanent, legible, and of uniform width. The capacity lines shall be perpendicular to the vertical axis of the flask. The inside and exterior surfaces of the flask shall be smooth except for impressions from graduation marking, identification marking (see 3.4), and seams from mold marks in the fabrication of the flask.

3.3 Performance. The flask shall not wobble when placed on a flat surface. The inner surface at the top of the neck shall accommodate the following standard size tapered stoppers conforming to ZZ-S-751, Type A and Grade I (see table I):

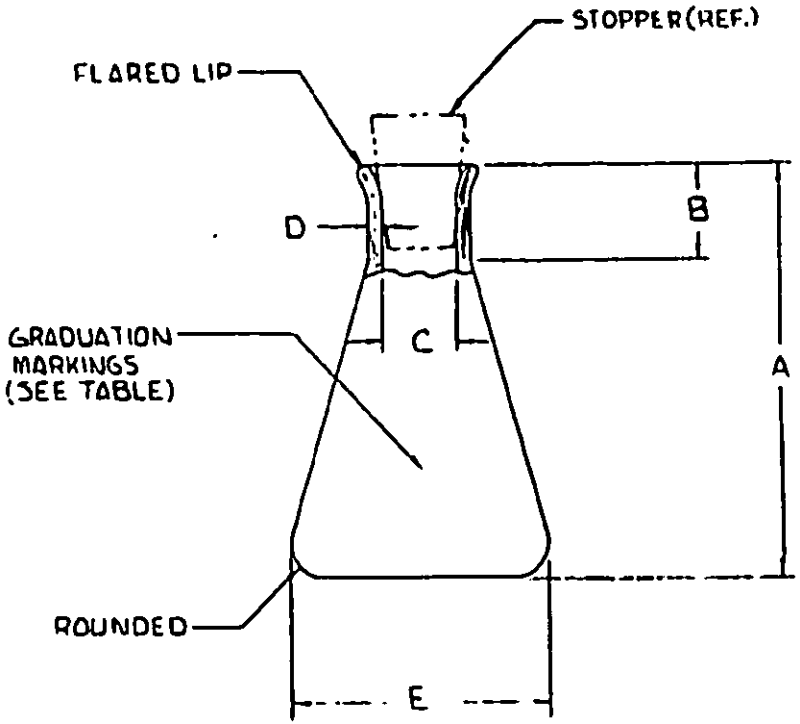
Table I. Stoppers for Erlenmeyer Flasks

Erlenmeyer flask	Stopper number
Size 1	#1
Size 2	#6
Size 3	#9

The stoppers indicated above are not required to be supplied with the flask.

When tested per 4.4.1, the volumetric accuracy of the graduation markings on the flask shall be within ± 5% for the top graduation (see 1.2) and within ± 10% for the other graduations. The flask shall be autoclavable; and when tested per 4.4.2, the flask shall not soften, deform, crack or become permanently opaque.

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SIZE	DIMENSIONS					GRADUATION MARKING	
	A	B	C	D	E	GRADUATION INCREMENTS	NUMBERED GRADUATIONS
1	3 1/8	5/8	1/2 TO 1 1/16	1/16	2 1/8	10 ML INCREMENTS FROM 20 TO 50 ML	20, 30, 40 AND 50
2	5 3/8	1 1/4	1/8 TO 1 1/16	1/8	3 3/8	25 ML INCREMENTS FROM 50 TO 250 ML	50, 100, 150, 200 AND 250
3	8 9/16	1 5/8	1 1/4 TO 1 1/2	1/8	5 1/4	50 ML INCREMENTS FROM 200 TO 1000 ML	100 ML INCREMENTS FROM 200 TO 1000 ML.
	OVERALL HEIGHT (INCHES MAXIMUM)	LENGTH OF NECK (INCHES MINIMUM)	INSIDE D. OF NECK (MIN. TO MAX.) (INCHES)	WALL THICKNESS (INCHES MINIMUM)	EXTREME OUTSIDE D. AT BOTTOM (INCHES MAXIMUM)		

FIGURE I

I.I.L. F-36747A(USA-DN)

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3.4 Identification marking. Each Erlenmeyer flask shall be legibly and permanently impressed with the name or registered trademark of the manufacturer.

3.5 Workmanship. Workmanship shall be first class throughout. The Erlenmeyer flasks shall be free of deformation, flask, cracks, sharp edges, protrusions, discoloration, opaqueness, crazing, foreign matter or other defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the polycarbonate material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock No. (NSN), the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for examination purposes shall be the individual flask.

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Table II. Sampling Criteria for Examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. The unit of product for testing shall be the individual flask.

Table III. Sampling Criteria for Tests

Characteristic	Requirement	Test procedure	Inspection level	AQL
Volumetric accuracy	3.3	4.4.1	S-2	*
Autoclavability	3.3	4.4.2	S-2	*

*Acceptance number shall be zero regardless of lot size.

4.3 Examination. The flasks shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in Table IV.

Table IV. Classification of Defects

Categories	Defects*
Major	
101	Flask not free from any deformation.
102	Flask not free from oil or grease smears on either inside or outside.
103	Graduation markings not clearly defined or not easily read.
104	Flask not free from water drops or dried stains.
105	Exterior of flask not free of burrs, sharp protrusions, or sharp abrasions.

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Table IV. Classification of Defects (Cont'd.)

106	Flask not free of cuts, tears, or cracks.
107	Flask not stable when placed on a flat surface.
108	Sides of flask not free of opaqueness.
109	Flask not capable of being fitted with stopper specified in 3.3.
Minor:	
201	Manufacturer's name or trademark not present, not clear or not distinct.
202	Flask not free of flash.
203	Flask not free of loose dirt or loose foreign particles on either inside or outside.
204	Flask not free of imbedded foreign particles.
205	Sides of flask not smooth on either inside or outside.
206	Sides of flask not free of crazing.

* Examination shall not be restricted to the above classified defects.

4.3.2 Dimensional examination. The Erlenmeyer flask shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

Table V. Examination of Preparation for Delivery

Examine	Defect
Contents	Not as specified
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Packaging and packing material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

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4.3.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with the specification requirements.

4.4.1 Volumetric accuracy. The flask under test shall be thoroughly clean and dry, and shall be tested by using distilled water. The test shall be conducted at a room temperature of 25° Centigrade (C.) ± 5°. Before starting the test, the test flask and distilled water shall stand at room temperature for 30 minutes. The room temperature shall be recorded. The accuracy of all numbered graduations indicated in the table of figure I shall be tested by either of the following methods:

Method A. Distilled water of known volume (to nearest 0.5 ml.) shall be delivered so that the test flask is filled to the capacity line under test. The distilled water shall be delivered from calibrated vessels and/or burets or pipets that yield a maximum overall accuracy of 1%. The volume of distilled water delivered shall be recorded. The accuracy of all graduations under test may be obtained by cumulative readings.

Method B. The test flask shall first be weighed empty (to the nearest 0.5 gram); and then again weighed after being filled to the capacity line under test. The two weights shall be recorded. The difference between the two weights shall be recorded and shall be considered the weight of the water delivered. The density of the water, corresponding to the observed room temperature, shall be obtained from a suitable handbook. The volume of water delivered shall be calculated by using the following formula:

$$V = \frac{W_2 - W_1}{D}$$

Where: V = Calculated volumetric capacity (ml.).

W_1 = Weight of empty flask (grams).

W_2 = Weight of the filled flask (grams).

D = Density of water at observed temperature $(\frac{\text{grams}}{\text{ml.}})$

The accuracy of all graduations under test may be obtained by cumulative readings.

4.4.2 Autoclavability. The flask under test shall be autoclaved at 15 p.s.i. and 250° Fahrenheit (F.) for 30 minutes; and then cooled and examined for any evidence of deterioration or damage.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. Each flask, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table VI shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VI. Packaging Quantities

Size	Unit package	Intermediate package	Exterior container
1	1 unit	12 units	72 units
2	1 unit	6 units	48 units
3	1 unit	6 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

* 5.1.3.1 Unit package. Unit package shall be a box of appropriate size constructed in accordance with PPP-B-566, or PPP-B-676 or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

* 5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard boxes shall be closed by method I or II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

* 5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Erlenmeyer flasks covered by this specification are intended for use in general laboratory procedures for which steam sterilization may be required.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National stock number (NSN)
(c) Specify size flask required (see 1.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 This specification covers the following items which appear in the Federal Supply Catalog:

NSN	Item Identification
6640-00-010-8360	FLASK, ERLENMEYER, Graduated, Plastic, 50 ml.
6640-00-010-8361	FLASK, ERLENMEYER, Graduated, Plastic, 250 ml.
6640-00-010-8362	FLASK, ERLENMEYER, Graduated, Plastic, 1000 ml.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - MD
Navy - MS
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Preparing activity:
Defense Supply Agency - DM

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DOCUMENT IDENTIFIER AND TITLE

MIL-F-36747A(DSA-DM) - FLASK, ERLENMEYER, GRADUATED, PLASTIC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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